Electron tube — Solution

Y25 (2025) — English translation

A1 — 0.30 pts

Let the electron begin to move with zero initial speed from the point C. Show that in this case $x(t) = 0$.

At the initial moment, the electron acquires speed along the Oy axis, along the electric field. This leads to the emergence of a magnetic force along Oz and the appearance of a velocity component v_z. The magnetic force associated with v_z is parallel to the Oy axis. Thus, in the process of further movement, the forces acting on the electron lie in the Oyz plane, from which follows $x(t) = 0$.

A2 — 0.50 pts

For the described case, write down Newton's second law in projection on the y, z axes, from which you will obtain expressions for $\ddot{y}$, $\ddot{z}$. Expressions may contain $\dot{y}$, $\dot{z}$, y, z, U, I, as well as geometric characteristics of the system.

In a flat configuration, the electric field is uniform and equal to

$$E_y = \frac{\varphi_{\mathrm{C}} - \varphi_{\mathrm{A}}}{h} = -\frac{U}{h}.$$

Magnetic field wire wire for motion electron in plane Oyz in parallel Ox equal (for example, from the circulation theorem)

$$B_x = -\frac{\mu_0 I}{2\pi(y + r_0)}.$$

Taking into account the negative sign of the electron's charge, we obtain the equations of motion:

$$\begin{cases} m\ddot{y} = -eE_y - e\dot{z}B_x \\ m\ddot{z} = e\dot{y}B_x \end{cases}$$

Substituting expressions for the fields, we get the answer.

Answer:

$$\begin{cases} \ddot{y} = \dfrac{eU}{mh} + \dfrac{\mu_0 eI}{2\pi m(y + r_0)}\dot{z} & (1) \\ \ddot{z} = -\dfrac{\mu_0 eI}{2\pi m(y + r_0)}\dot{y} & (2) \end{cases}$$

A3 — 1.00 pts

Obtain a differential equation for y of the form

$$\ddot{y} = \alpha - \beta\frac{\ln\left(1 + \frac{y}{r_0}\right)}{1 + \frac{y}{r_0}},$$

where α, β – constant. Find α, β. The answer may contain all previously entered quantities and constants.

Multiplying the second equation by dt, we arrive at the relation

$$d\dot{z} = -\frac{\mu_0 eI}{2\pi m(y + r_0)}dy.$$

Integrate:

$$\dot{z} = -\frac{\mu_0 eI}{2\pi m}\int\frac{dy'}{y'+r_0} = -\frac{\mu_0 eI}{2\pi m}\ln\left(1+\frac{y}{r_0}\right)+C.$$

C accounting for initial condition $\dot{z}=0$ for $y=0$, then $C=0$, and

$$\dot{z} = -\frac{\mu_0 eI}{2\pi m}\ln\left(1+\frac{y}{r_0}\right).$$

Substituting the obtained expression into (1) gives

$$\ddot{y} = \frac{eU}{mh} - \frac{1}{r_0}\left(\frac{\mu_0 eI}{2\pi m}\right)^2\frac{\ln\left(1+\frac{y}{r_0}\right)}{1+\frac{y}{r_0}}. \;(3)$$

From here we get the answer.

Answer:

$$\alpha = \frac{eU}{mh}\beta = \frac{1}{r_0}\left(\frac{\mu_0 eI}{2\pi m}\right)^2$$

A4 **0.80 pts**

Find the maximum value of the heater current I_{cr} at which the electron in question reaches the anode.

After multiplying by $\dot{y}$, the equation (3) is transformed to the form

$$\frac{\dot{y}^2}{2} = \int\left(\frac{eU}{mh} - \frac{1}{r_0}\left(\frac{\mu_0 eI}{2\pi m}\right)^2\frac{\ln\left(1+\frac{y}{r_0}\right)}{1+\frac{y}{r_0}}\right)dy.$$

Integral in second term:

$$\int\frac{\ln\left(1+\frac{y}{r_0}\right)}{1+\frac{y}{r_0}}dy = r_0\int\ln\left(1+\frac{y}{r_0}\right)d\left(\ln\left(1+\frac{y}{r_0}\right)\right) = \frac{r_0}{2}\ln^2\left(1+\frac{y}{r_0}\right)+C.$$

Taking into account initial condition $\dot{y}=0$ for $y=0$, then $C=0$, and

$$\frac{\dot{y}^2}{2} = \frac{eUy}{mh} - \frac{1}{2}\left(\frac{\mu_0 eI}{2\pi m}\right)^2\ln^2\left(1+\frac{y}{r_0}\right).$$

Condition for the electron to reach the anode:

$$\dot{y}(h)=0, 0 = \frac{eU}{m} - \frac{1}{2}\left(\frac{\mu_0 eI_{\text{cr}}}{2\pi m}\right)^2\ln^2\left(1+\frac{h}{r_0}\right).$$

Hence we get answer for I_{cr}.

Answer:

$$I_{\text{cr}} = \sqrt{\frac{8Um}{e}}\frac{\pi}{\mu_0}\frac{1}{\ln\left(1+\frac{h}{r_0}\right)}$$

A5 **1.20 pts**

Let the electron start moving with zero initial speed from the point C in the cylindrical lamp configuration. In this case, find the maximum value of the heater current I_{cr} at which the electron reaches the anode.

Similar to the previous situation, it is shown that the movement will occur in the Oyz plane if the axes are introduced in the same way as in the figure for a plane configuration.

Method 1

We will consider the movement in coordinates (r, z), where r is the distance from the system axis to the point. Let's calculate the electric field using Gauss's theorem. Due to the absence of volumetric charges

$$E_r \cdot r = \text{const.} \quad \Rightarrow \quad E_r = \frac{A}{r}, U = -\int_{r_0}^{R_0} E_r dr = -A\ln\left(\frac{R_0}{r_0}\right), E_r = -\frac{U}{r\ln\left(\frac{R_0}{r_0}\right)}.$$

Magnetic field:

$$B_x = -\frac{\mu_0 I}{2\pi r}.$$

Then equations of motion look like following manner:

$$\begin{cases} \ddot{r} = -\dfrac{eU}{mr\ln\left(\frac{R_0}{r_0}\right)} + \dfrac{\mu_0 eI}{2\pi m(y+r_0)}\dot{z} \\ \ddot{z} = -\dfrac{\mu_0 eI}{2\pi m(y+r_0)}\dot{r} \end{cases}$$

In an analogous manner, after integrating the second equation, substituting into the first and integrating again, we obtain the equation for I_{cr}:

$$\frac{eU}{m} - \frac{1}{2}\left(\frac{\mu_0 eI_{\text{cr}}}{2\pi m}\right)^2 \ln^2\left(\frac{R_0}{r_0}\right).$$

From here we get the answer.

Method 2

Let's consider the equations of motion from point A2, without substituting the specific values of E_y and B_x:

$$\begin{cases} \ddot{y} = -\dfrac{e}{m}\left(E_y(y) + \dot{z}B_x(y)\right) \\ \ddot{z} = \dfrac{e}{m}\dot{y}B_x(y) \end{cases}.$$

Taking into account initial condition from second equation obtain:

$$\dot{z} = \frac{e}{m}\int_0^y B_x(\xi)d\xi.$$

Substitute in first equation:

$$\ddot{y} = -\frac{e}{m}\left(E_y(y) + \frac{e}{m}B_x(y)\int_0^y B_x(\xi)d\xi\right).$$

Multiply by $\dot{y}$ and integrate taking into account initial condition:

$$\frac{\dot{y}^2}{2} = -\frac{e}{m}\int_0^y E_y(\eta)d\eta - \frac{e^2}{m^2}\int_0^y B_x(\eta)d\eta\int_0^\eta B_x(\xi)d\xi = = \frac{e}{m}(\varphi(y) - \varphi(0)) - \frac{e^2}{m^2}\int_0^y B_x(\eta)d\eta\int_0^\eta B_x(\xi)d\xi.$$

Setting $\dot{y}(h) = 0$ gives the condition for the electron to reach the anode:

$$\frac{e^2}{m^2}\int_0^h B_x(\eta)d\eta\int_0^\eta B_x(\xi)d\xi = \varphi(h) - \varphi(0) = U.$$

Thus, condition does not depend on the distribution of the field E_y, but only on the potential difference U. Since the distribution of the magnetic field in the cylindrical and in the flat system coincide, the answer for I_{cr} (with the substitution $R_0 = r_0 + h$) remains the same.

Note: It can be shown that

$$\int_0^h B_x(\eta)d\eta \int_0^\eta B_x(\xi)d\xi = \frac{1}{2}\left(\int_0^h B_x(\xi)d\xi\right)^2.$$

Thus, the condition for the electron to reach the anode is:

$$\int_0^h B_x(\xi)d\xi = \sqrt{\frac{2mU}{e}}.$$

An interesting fact is that the obtained condition depends only on the values of the definite integrals of the electric and magnetic fields over y, and not on the specific field distributions.

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Thus, the condition for the electron to reach the anode is:

$$\int_0^h B_x(\xi)d\xi = \sqrt{\frac{2mU}{e}}.$$

An interesting fact is that the found condition depends only on the values of certain integrals of the electric and magnetic field over y, but not on the specific field distribution.

Answer:

$$I_{\text{cr}} = \sqrt{\frac{8Um}{e}}\frac{\pi}{\mu_0}\frac{1}{\ln\left(\frac{R_0}{r_0}\right)}$$

B1 — 0.30 pts

Derive a differential equation relating $\varphi(y)$ and $\rho(y)$.

Method 1

Let's write Gauss's theorem for two planes y and $y+dy$:

$$\left(E(y+dy) - E(y)\right)S = \frac{\rho(y)}{\varepsilon_0}Sdy, E'(y) = \frac{\rho(y)}{\varepsilon_0}.$$

Taking into account relation

$$E(y) = -\varphi'(y)$$

we get the answer.

Method 2

Gauss's theorem in differential form:

$$\frac{\rho}{\varepsilon_0} = \operatorname{div}\vec{E}.$$

Relation E and φ:

$$E = -\operatorname{grad}\varphi.$$

Then

$$\frac{\rho}{\varepsilon_0} = -\operatorname{div}\operatorname{grad}\varphi = -\Delta\varphi,$$

here $\Delta = \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2}$ is the Laplace operator (Laplacian), the equation itself is called the Laplace equation. In the one-dimensional case this transforms to the form

Answer:

$$\varphi''(y) = -\frac{\rho(y)}{\varepsilon_0}$$

B2 — 0.20 pts

Let's assume that $U > 0$. Find the speed of the electron $v(y)$ emitted from the cathode at a distance y from the cathode. Express your answer using $\varphi(y)$.

Let us write down the law of conservation of energy for the electron:

$$\frac{mv^2(y)}{2} - e\varphi(y) = 0.$$

From this follows the answer.

Answer:

$$v = \sqrt{\frac{2e\varphi(y)}{m}}$$

B3 — 0.30 pts

Express the total current I flowing through the section $y = const$. Express your answer in terms of $\rho(y)$, $v(y)$ and the geometric characteristics of the system. Consider the direction of current from the anode to the cathode to be positive.

Taking into account the indicated positive direction of current from the anode to the cathode

Answer:

$$I = -\rho(y)v(y)S$$

B4 — 0.40 pts

Obtain a differential equation on $\varphi(y)$ containing the function $\varphi(y)$ and its derivatives, as well as I and the geometric characteristics of the system.

Taking into account the previous points:

$$\rho(y) = -\frac{I}{Sv(y)} = -\frac{I}{S}\sqrt{\frac{m}{2e\varphi(y)}}.$$

Substitution $\rho(y)$ into the equation of point B1 leads to the answer

Answer:

$$\varphi''(y) = \frac{I}{\varepsilon_0 S}\sqrt{\frac{m}{2e\varphi(y)}}$$

B5 — 0.80 pts

Get addicted $\varphi(y)$. Express your answer in terms of I and the geometric characteristics of the system.

Multiplying both sides of the equation of item B4 by $\varphi'(y)$ and integrating taking into account $\varphi'(0) = 0$, we arrive at the equation

$$(\varphi')^2 = \frac{I}{\varepsilon_0 S}\sqrt{\frac{8m}{e}}\sqrt{\varphi}.$$

The variables in it separate:

$$\varphi^{-1/4} d\varphi = \left(\frac{I}{\varepsilon_0 S}\right)^{1/2}\left(\frac{8m}{e}\right)^{1/4} dy.$$

Apply initial condition $\varphi(0) = 0$, obtain

$$\frac{4}{3}\varphi^{3/4} = \left(\frac{I}{\varepsilon_0 S}\right)^{1/2}\left(\frac{8m}{e}\right)^{1/4} y.$$

Transforming the resulting expression, we arrive at the answer.

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$$\varphi^{-1/4} d\varphi = \left(\frac{I}{\varepsilon_0 S}\right)^{1/2} \left(\frac{8m}{e}\right)^{1/4} dy.$$

Applying the initial condition $\varphi(0) = 0$, we obtain

$$\frac{4}{3}\varphi^{3/4} = \left(\frac{I}{\varepsilon_0 S}\right)^{1/2} \left(\frac{8m}{e}\right)^{1/4} y.$$

Transforming the resulting expression, we arrive at the answer.

Answer:

$$\varphi(y) = \left(\frac{3y}{4}\right)^{4/3} \left(\frac{I}{\varepsilon_0 S}\right)^{2/3} \left(\frac{8m}{e}\right)^{1/3}$$

B6 **0.80 pts**

The current-voltage characteristic of the lamp at $U > 0$ can be expressed by the function

$$I(U) = GU^{\gamma}.$$

Determine coefficient G, γ. Express the answer through the geometric characteristics of the system. Quantity G is called the perveance of the lamp.

The value G is called the lamp perveance.

The voltage across the lamp is equal to the potential at point $y = h$:

$$U = \varphi(h) = \left(\frac{3h}{4}\right)^{4/3} \left(\frac{I}{\varepsilon_0 S}\right)^{2/3} \left(\frac{8m}{e}\right)^{1/3}.$$

Express current:

$$I(U) = \varepsilon_0 S \left(\frac{4}{3h}\right)^2 \left(\frac{e}{8m}\right)^{1/2} U^{3/2}.$$

Then

Answer:

$$G = \varepsilon_0 S \left(\frac{4}{3h}\right)^2 \left(\frac{e}{8m}\right)^{1/2}$$

$$\gamma = \frac{3}{2}$$

B7 **0.50 pts**

Get addicted $\rho(y)$. Express your answer in terms of I and the geometric characteristics of the system.

Using the results of points B1 or B4, we express $\rho(y)$ in terms of $\varphi(y)$:

$$\rho(y) = -\varepsilon_0 \cdot \varphi''(y) = -\frac{I}{S}\sqrt{\frac{m}{2e\varphi(y)}}.$$

Substitute find $\varphi(y)$, we arrive at the answer

Answer:

$$\rho(y) = -\left(\frac{I}{3S}\right)^{2/3} \left(\frac{2m\varepsilon_0}{e}\right)^{1/3} \cdot y^{-2/3}$$

B8 0.20 pts

Obtain the current-voltage characteristic of lamp $I(U)$ at $U<0$.

At $U<0$ the field is directed along the y axis, and the electrons emitted from the cathode do not move towards the anode, which means $I=0$.

Answer:

$$I(U)=0$$

B9 0.60 pts

Obtain a differential equation for $\varphi(r)$ containing the function $\varphi(r)$ and its derivatives, as well as the total current in the lamp I and the geometric characteristics of the system.

The current flowing in a cylindrical configuration is

$$I=-2\pi r L\rho(r)v(r).$$

Taking into account expression for velocity obtain

$$\rho(r)=-\frac{I}{2\pi r L}\sqrt{\frac{m}{2e\varphi(r)}}$$

Method 1

Consider cylindrical surfaces of length L and radii r and $r+dr$. Gauss's theorem in this case will be written in the form:

$$2\pi L\left(E(r+dr)(r+dr)-E(r)r\right)=\frac{2\pi L r\rho(r)dr}{\varepsilon_0},\frac{1}{r}\frac{d}{dr}\left(rE(r)\right)=\frac{\rho(r)}{\varepsilon_0}=-\frac{I}{2\pi r L\varepsilon_0}\sqrt{\frac{m}{2e\varphi(r)}}.$$

Use $E(r)=-\varphi(r)$, we arrive at the equation for the potential.

Method 2

In method 2 of point B1,

$$\Delta\varphi=-\frac{\rho}{\varepsilon_0}.$$

The Laplace operator in cylindrical coordinates:

$$\Delta\varphi=\frac{1}{r}\frac{\partial}{\partial r}\left(r\frac{\partial\varphi}{\partial r}\right)+\frac{1}{r^2}\frac{\partial^2\varphi}{\partial\theta^2}+\frac{\partial^2\varphi}{\partial z^2}.$$

due to the axial symmetry of the system and $E_z=0$, the last two terms equal zero. After substitution ρ through I was obtained, we get the answer.

Answer:

$$r\varphi''(r)+\varphi'(r)=\frac{I}{2\pi L\varepsilon_0}\sqrt{\frac{m}{2e\varphi(r)}}$$

C1 0.90 pts

Kirchhoff's law for the RLC circuit leads to a differential equation of the form

$$\ddot{U}_C+\zeta(1+\eta U_C^2)\dot{U}_C+\chi U_C=0.$$

Express ζ, η, χ through R, L, C, M, λ, μ.

The current in the RLC circuit is related to the capacitor voltage:

$$I=C\dot{U}_{\mathrm{C}}.$$

Then for RLC-circuit is written Kirchhoff's law:

$$M\frac{dI_{\mathrm{A}}}{dt}=U_{\mathrm{C}}+RC\dot{U}_{\mathrm{C}}+LC\ddot{U}_{\mathrm{C}}\ (1)$$

(with accuracy to sign M). Use expression for $I_\text{A}(U_\text{C})$, obtain

$$\frac{dI_\text{A}}{dt} = \left(\lambda - 3\mu U_\text{C}^2\right)\dot{U}_\text{C}. \ (2)$$

After substitution (2) in (1) and dividing by LC, we arrive at the equation

$$\ddot{U}_\text{C} + \left(\frac{R}{L} - \frac{\lambda M}{LC} + \frac{3\mu M}{LC}U_\text{C}^2\right)\dot{U}_\text{C} + \frac{U_\text{C}}{LC} = 0.$$

Comparing it with the data in the condition, we get the answer.

Answer:

$$\zeta = \frac{R}{L} - \frac{\lambda M}{LC}\eta = \frac{3\mu M}{RC - \lambda M}\chi = \frac{1}{LC}$$

C2 1.20 pts

Let U_C deviate from zero by a small amount. Obtain a condition for the parameters ζ, η, χ under which the oscillations build up. Also express the resulting condition in terms of R, L, C, M, λ, μ.

For small U_C we can neglect the $\sim U_\text{C}^2$ term in the equation and write it in the form:

$$\ddot{U}_\text{C} + \zeta\dot{U}_\text{C} + \chi U_\text{C} = 0.$$

Method 1 (mathematical)
Let's introduce the notation:

$$\zeta = 2\gamma, \qquad \chi = \omega_0^2.$$

As is known, solution such differential equation have different form in dependence from parameter γ, ω_0.
$\gamma^2 < \omega_0^2$

$$U_\text{C}(t) = e^{-\gamma t}\left(A\cos\left(\sqrt{\omega_0^2 - \gamma^2}\cdot t\right) + B\sin\left(\sqrt{\omega_0^2 - \gamma^2}\cdot t\right)\right).$$

$\gamma^2 = \omega_0^2$

$$U_\text{C}(t) = e^{-\gamma t}(A + Bt).$$

$\gamma^2 > \omega_0^2$

$$U_\text{C}(t) = e^{-\gamma t}\left(A\exp\left(\sqrt{\gamma^2 - \omega_0^2}\cdot t\right) + B\exp\left(-\sqrt{\gamma^2 - \omega_0^2}\cdot t\right)\right).$$

The second case does not have physical meaning, since it requires the exact fulfillment of the condition $\gamma^2 = \omega_0^2$. in the remaining cases, as can be seen, build-up of oscillations is possible only under the condition $\gamma < 0$, that is $\zeta < 0$.

As is known, the solution to such a differential equation has a different form depending on the parameters γ, ω_0.

The second case has no physical meaning, since the exact fulfillment of the condition $\gamma^2 = \omega_0^2$ is required. In other cases, as can be seen, swinging is possible only under the condition $\gamma < 0$, that is, $\zeta < 0$.

In the third case, this does not lead to exponential growth of U_C, since the resulting linear equation is valid only for small U_C. A characteristic phase diagram of the exact solution is shown in the figure.

Method 2 (energy)

Energy in the RLC circuit:

$$W = \frac{CU_\text{C}^2}{2} + \frac{LC^2\dot{U}_\text{C}^2}{2}.$$

Differentiate by time:

$$\dot{W} = CU_\text{C}\dot{U}_\text{C} + LC^2\dot{U}_\text{C}\ddot{U}_\text{C} = C\dot{U}_\text{C}(U_\text{C} + LC\ddot{U}_\text{C}).$$

Substitute $\ddot{U}_\text{C}$ from the original equation, express it through $\dot{U}_\text{C}$, U_C. After transformation, we arrive at expression

$$\dot{W} = C\dot{U}_\text{C}^2(M\lambda - RC - 3\mu MU_\text{C}^2).$$

For small U_{C}:

$$\dot{W} \simeq C\dot{U}_{\mathrm{C}}^{2}(M\lambda - RC).$$

For build-up of oscillations oscillation necessarily $\dot{W} > 0$, which is equivalent

Answer:

$$\zeta < 0 RC < \lambda M$$